上海创智学院
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Training Uzi: PPO for Lane-Dominant 1v1 Honor of Kings Agents

Team Uzi

Feature semantics, pointer policy, staged curriculum, and verified lane pressure

Abstract

We train Uzi, a PPO-based 1v1 Honor of Kings agent, under a lane-dominant strategy. The system converts raw frame state into canonical entity tokens, repairs target/action semantics, and uses a pointer-style policy to choose buttons and entity targets. The final agent emphasizes reliable lane pressure: last hits, safe trades, valid targets, tower conversion, and external match win rate.

Motivation

- Raw game state mixes heroes, towers, minions, bullets, cakes, and transient visibility.
- Action buttons and entity targets must agree; otherwise the policy learns invalid choices.
- Sparse win/loss feedback is delayed, so reward shaping must expose lane pressure early.
- Negative findings, including recall and hero-specific timing, are kept as design evidence.

Method

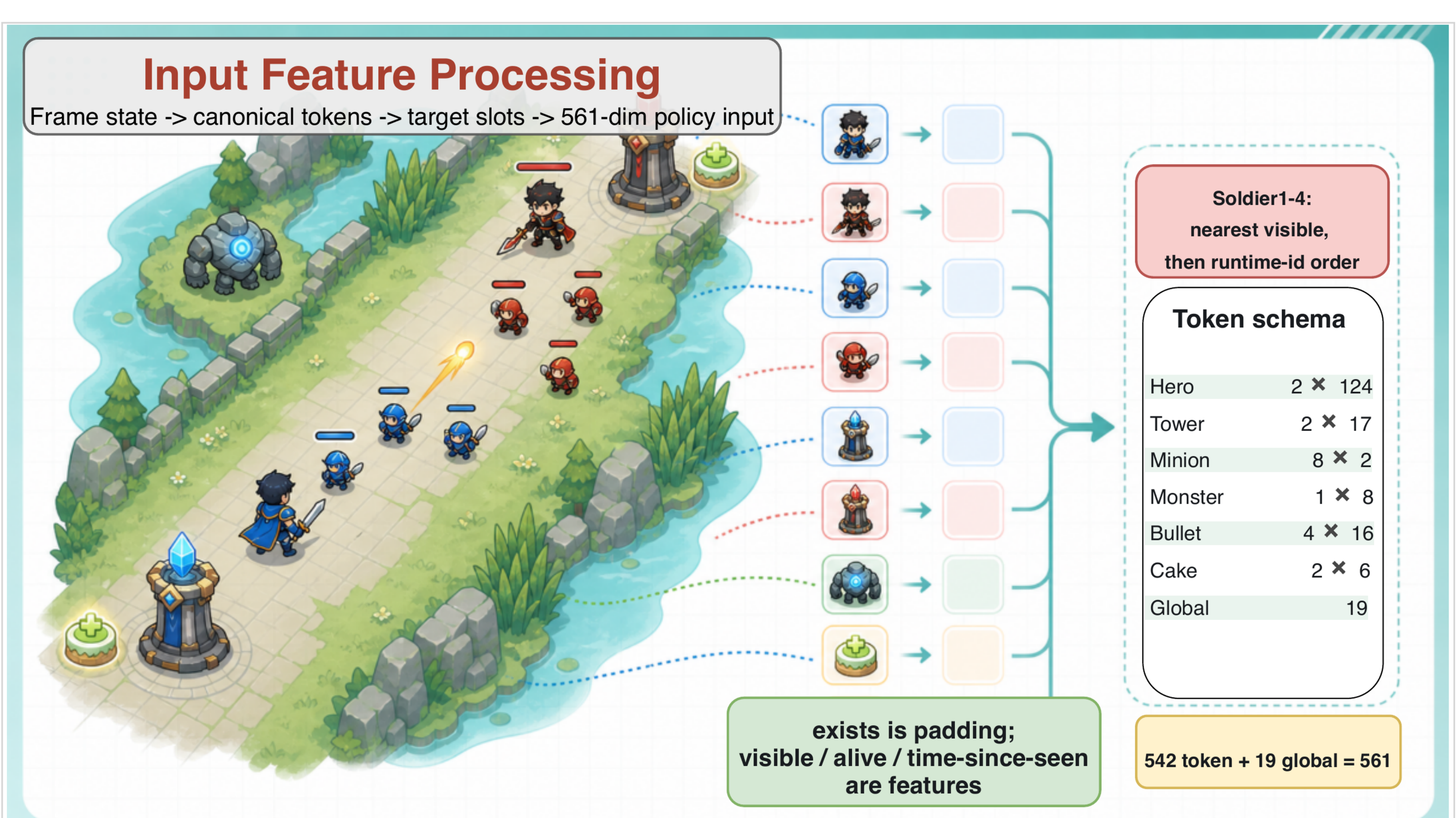
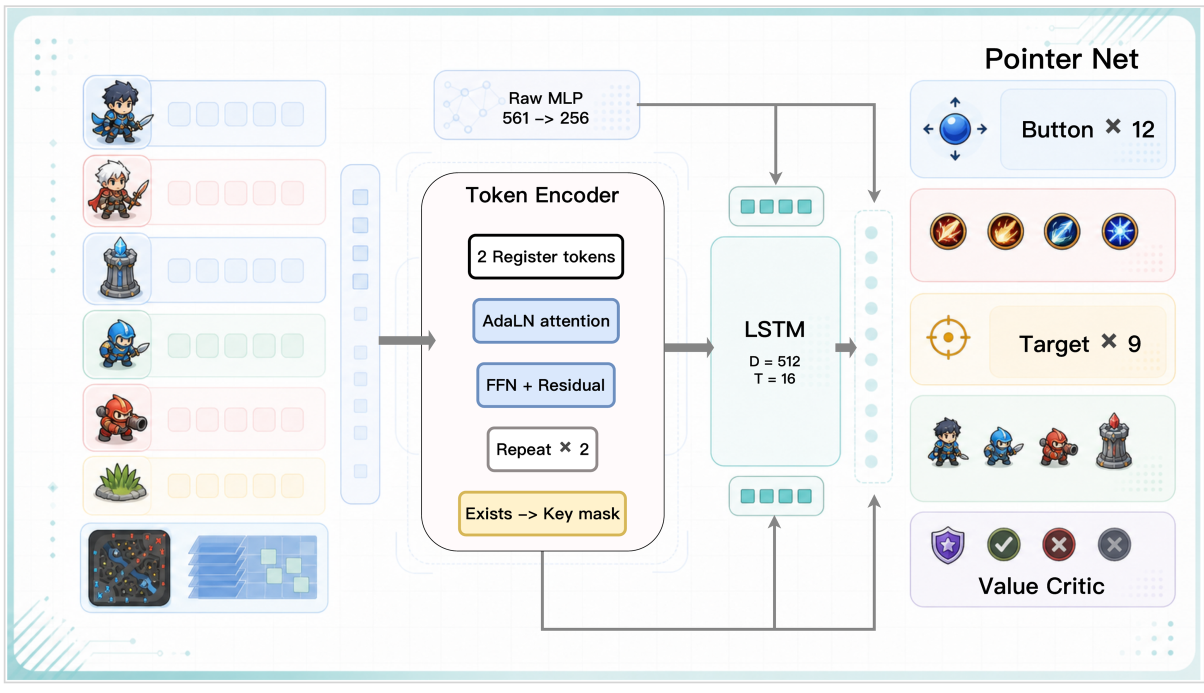


Figure 1: raw frame state is canonicalized into stable tokens and target slots, producing a 561-dim policy input.

The policy then encodes tokens, carries temporal state through an LSTM, and predicts button, target, and value heads. Target logits are conditioned on the selected button, matching the game API rather than treating actions as independent labels.



Policy architecture: token encoder and LSTM memory feed pointer heads for buttons, targets, and value.

Experiment

Setup:

- Environment: 1v1 Honor of Kings lane task.
- Policy: token encoder, LSTM memory, pointer button and target heads.
- Training: PPO with curriculum stages, self-play/common-AI mix, and reward shaping.
- Evidence: course leaderboard, local logs, real frame probes, and synthetic history probes.

Main Result:

Uzi
Rank #3
131 / 67
A wins / B wins over 198 games
66.2%
aggregate win rate

Uzi finished Rank #3 with 131 wins and 67 losses over 198 games, for a 66.2% aggregate win rate.

Training Curriculum:

Figure 2: staged training moves from basic control to lane aggression and final win conversion.

Diagnostics:

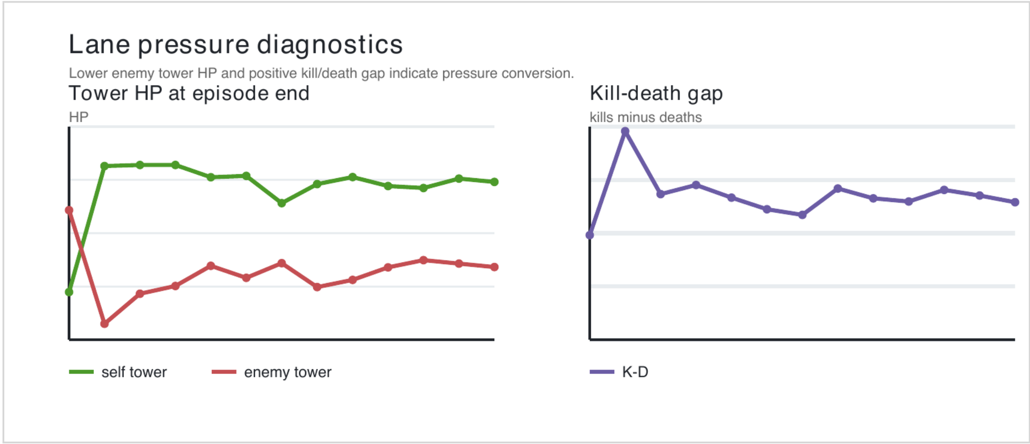
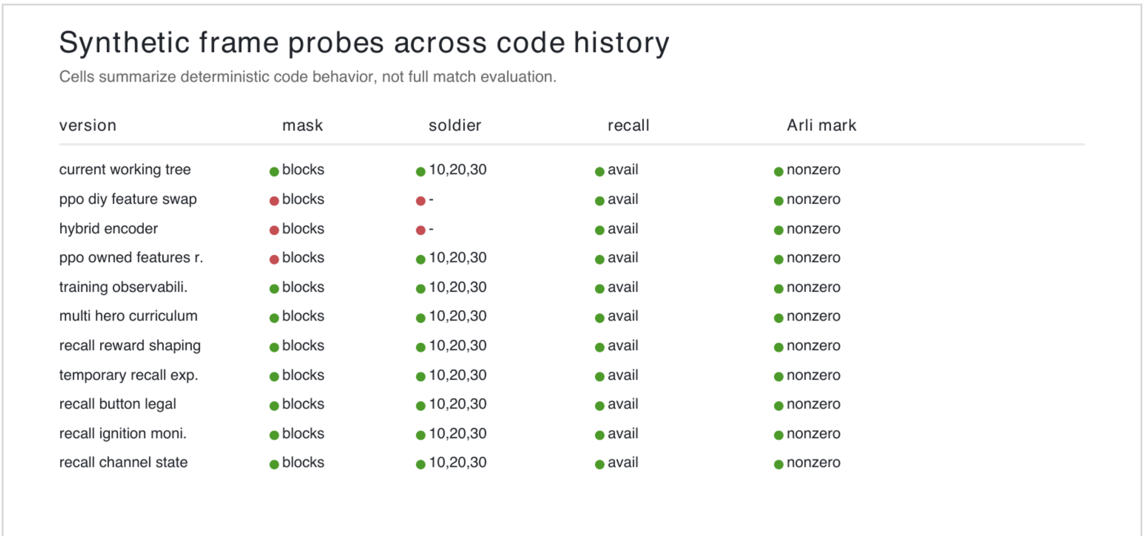


Figure 3: lower enemy tower HP and positive kill-death gap indicate pressure conversion.



Synthetic frame probes across code history

Cells summarize deterministic code behavior, not full match evaluation.

version	mask	soldier	recall	Arli mark
current working tree	blocks	10,20,30	avail	nonzero
ppo diy feature swap	blocks	-	avail	nonzero
hybrid encoder	blocks	-	avail	nonzero
ppo owned features r.	blocks	10,20,30	avail	nonzero
training observabili.	blocks	10,20,30	avail	nonzero
multi hero curriculum	blocks	10,20,30	avail	nonzero
recall reward shaping	blocks	10,20,30	avail	nonzero
temporary recall exp.	blocks	10,20,30	avail	nonzero
recall button legal	blocks	10,20,30	avail	nonzero
recall ignition moni.	blocks	10,20,30	avail	nonzero
recall channel state	blocks	10,20,30	avail	nonzero

Figure 4: probe matrix tracks semantic fixes and remaining limitations across code history.